

Course 6044C

Semester VI

Theory

4 Credits

# Drone Technology

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes

**Electronics Engineering**

Contact hours

**60**

Teaching scheme

**3:1:0:0**

Credits

**4**

CIE / ESE

**Theory CIE 40 / Theory ESE 60**

Self-learning

**5.5 h/week**

## CURRICULUM INTEGRITY

# Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 6044C as Drone Technology in Semester VI for Electronics Engineering. The official SITTTTR detailed course page was checked at the indexed link and returned "Requested file not found." With the user's approved public-source approach, this handbook is transparently based on NPTEL IISc Bangalore's Drone Systems and Control course and polytechnic drone technology curricula. It does not claim to reproduce official REV2026 detailed-syllabus wording.

**Source treatment:** Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. Drone designs, flight plans, pilot operations and maintenance procedures must be based on approved engineering plans, certified aviation data, current DGCA regulations and competent professional review.

**Verified source note:** Verified sources support UAV classifications, aerodynamics, drone components, and DGCA rules. The four-module sequence is a learning path, not an asserted official module split.

## COURSE DASHBOARD

# What You Are Expected to Learn

**60**

Contact hours

**4**

Credits

**Theory CIE 40**

CIE

**Theory ESE 60**

ESE

## Course introduction

Drone Technology (Unmanned Aerial Vehicles - UAV) studies the design, operation and applications of pilotless aircraft systems. It develops the ability to analyze flight aerodynamics

and dynamics, evaluate drone components like flight controllers and sensors, understand DGCA regulations and flight planning, and coordinate drone-based solutions while respecting the technical, legal and safety boundaries of electronic and aerospace engineering.

**Malayalam support:** ് handbook ് syllabus topic ്; ് principle, diagram/procedure, application, common mistake ് ്.

## Prerequisite foundation

Basic electronics, microcontrollers, sensors, wireless communication and basic physics (mechanics).

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

### Teaching and assessment scheme

| L | T | P | S | Self-learning/week | Contact hours | Credits | CIE           | ESE           |
|---|---|---|---|--------------------|---------------|---------|---------------|---------------|
| 3 | 1 | 0 | 0 | 5.5 h/week         | 60            | 4       | Theory CIE 40 | Theory ESE 60 |

## STUDY METHOD

# How to Use This Handbook

## 1 • Understand

Read terms, conditions and purpose; make a short definition in your own words.

## 2 • Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

## 3 • Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

## 4 • Revise

Use questions, quick cards and common-mistake notes before examination.

### OFFICIAL STATEMENTS

## Objectives and Course Outcomes

### Official course objectives

1. Explain the fundamentals of UAVs, drone classifications and basic aerodynamics.
2. Analyze the hardware components of a drone including flight controllers, motors and sensors.
3. Evaluate the flight dynamics, control systems and wireless communication for drones.
4. Explain the systematic DGCA regulations, flight safety and maintenance of drone systems.

### Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

**Malayalam support:** CO ഉള്ളത് exam-ൽ ഉപയോഗിക്കാൻ ഉപയോഗിക്കാൻ ഉപയോഗിക്കാൻ. Define, explain, apply ഉപയോഗിക്കാൻ action words ഉപയോഗിക്കാൻ.

### Official Course Outcomes

| CO  | Official outcome                                                          | Bloom level |
|-----|---------------------------------------------------------------------------|-------------|
| CO1 | Explain the fundamentals of UAVs, aerodynamics and drone classifications. | Understand  |
| CO2 | Analyze the selection and operation of drone hardware components.         | Apply       |
| CO3 | Analyze flight dynamics, control algorithms and communication links.      | Apply       |
| CO4 | Explain DGCA regulations, flight safety protocols and drone maintenance.  | Understand  |

### Official CO-PO articulation matrix

| CO  | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7-PO11 |
|-----|-----|-----|-----|-----|-----|-----|----------|
| CO1 | 3   | 2   | 2   | 2   | 2   | -   | -        |
| CO2 | 3   | 2   | 2   | 2   | 2   | -   | -        |
| CO3 | 3   | 2   | 2   | 2   | 2   | -   | -        |
| CO4 | 3   | 2   | 2   | 2   | 2   | -   | -        |

### SYLLABUS COVERAGE

## Curriculum Coverage Map

### All official major topics mapped to handbook chapters

| Module | Title                                 | Official major topics                                                                                                                                                               | Hours            | CO  | Handbook section         |
|--------|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-----|--------------------------|
| 1      | Introduction to UAVs and Aerodynamics | Definition of UAV/Drone; History and Evolution; Classifications: Nano, Micro, Small, Medium, Large; Aerodynamics: Lift, Weight, Thrust, Drag; Multi-rotor vs Fixed-wing.            | Approx. 14 hours | CO1 | <a href="#">Module 1</a> |
| 2      | Drone Hardware and Components         | Flight Controller (FC): Architecture and Sensors (IMU, Barometer, GPS); Electronic Speed Controllers (ESC); BLDC Motors and Propellers; Battery technology (LiPo); Frame materials. | Approx. 16 hours | CO2 | <a href="#">Module 2</a> |
| 3      | Flight Dynamics and Control           | 6-DOF Dynamics: Roll, Pitch, Yaw; Control algorithms: PID control basics; Wireless communication: Radio control (RC), Telemetry links; FPV (First Person View) systems.             | Approx. 16 hours | CO3 | <a href="#">Module 3</a> |
| 4      | Regulations, Safety and Maintenance   | DGCA Regulations: Digital Sky platform, UIN, Remote Pilot License; Airspace zones: Green, Yellow, Red; Pre-flight and Post-flight checklists; Maintenance and Troubleshooting.      | Approx. 14 hours | CO4 | <a href="#">Module 4</a> |

### LEARNING ROADMAP

## How the Modules Connect

### Introduction to UAVs and Aerodynamics

Definition of UAV/Drone; History and Evolution; Classifications: Nano, Micro, Small, Medium, Large; Aerodynamics: Lift, Weight, Thrust, Drag; Multi-rotor vs Fixed-wing.

CO1 · Approx. 14 hours

### Drone Hardware and Components

Flight Controller (FC): Architecture and Sensors (IMU, Barometer, GPS); Electronic Speed Controllers (ESC); BLDC Motors and Propellers; Battery technology (LiPo); Frame materials.

CO2 · Approx. 16 hours

### Flight Dynamics and Control

6-DOF Dynamics: Roll, Pitch, Yaw; Control algorithms: PID control basics; Wireless communication: Radio control (RC), Telemetry links; FPV (First Person View) systems.

CO3 · Approx. 16 hours

### Regulations, Safety and Maintenance

DGCA Regulations: Digital Sky platform, UIN, Remote Pilot License; Airspace zones: Green, Yellow, Red; Pre-flight and Post-flight checklists; Maintenance and Troubleshooting.

CO4 · Approx. 14 hours

## MODULE 1

# Introduction to UAVs and Aerodynamics

Definition of UAV/Drone; History and Evolution; Classifications: Nano, Micro, Small, Medium, Large; Aerodynamics: Lift, Weight, Thrust, Drag; Multi-rotor vs Fixed-wing.

Approx. 14 hours

CO1

Understand · Apply

### Learning goals

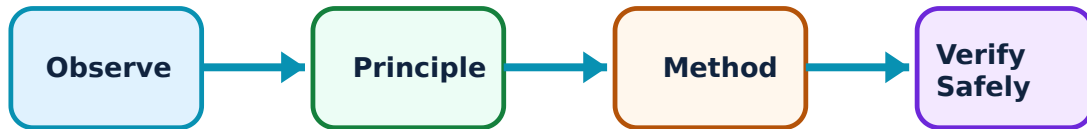
Identify core terms, explain the principle and complete the stated application or practical outcome.

### Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

## Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Introduction to UAVs and Aerodynamics: identify the system, state its principle, follow the correct method and verify the result safely.

### 1. Drone Fundamentals–

A Drone or Unmanned Aerial Vehicle (UAV) is an aircraft without a human pilot on board. Drones are classified by weight (e.g., Nano < 250g) and configuration. Multi-rotors (Quadcopters) are popular for their VTOL (Vertical Take-Off and Landing) capability, while Fixed-wing drones offer better efficiency for long-range tasks.

#### Study and application method

List the 'DGCA Weight Classifications' for drones; explain the four 'Aerodynamic Forces' acting on a drone in flight.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

#### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

## How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** List the 'DGCA Weight Classifications' for drones; explain the four 'Aerodynamic Forces' acting on a drone in flight.

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

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**Common mistake / precaution:** Improper classification can lead to legal violations. Do not operate drones without authorised understanding of weight limits and corresponding regulatory requirements.

## 2. Aerodynamics and Flight Basics–

Lift is generated by pressure differences (Bernoulli's principle). Weight is the force of gravity. Thrust is provided by motors/propellers. Drag is air resistance. Stable flight requires a balance of these forces. Control is achieved by varying motor speeds in multi-rotors.

### Study and application method

Describe how a 'Quadcopter' achieves 'Yaw' and 'Pitch' movements conceptually; compare 'Multi-rotor' and 'Fixed-wing' drones.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.



## How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** Describe how a 'Quadcopter' achieves 'Yaw' and 'Pitch' movements conceptually; compare 'Multi-rotor' and 'Fixed-wing' drones.

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**Common mistake / precaution:** Aerodynamic instability can cause crashes. Do not fly in high-wind conditions without authorised analysis of the drone's wind-resistance rating.

**Module 1 revision cue:** Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

## MODULE 2

# Drone Hardware and Components

Flight Controller (FC): Architecture and Sensors (IMU, Barometer, GPS); Electronic Speed Controllers (ESC); BLDC Motors and Propellers; Battery technology (LiPo); Frame materials.

Approx. 16 hours

CO2

Understand · Apply

## Learning goals

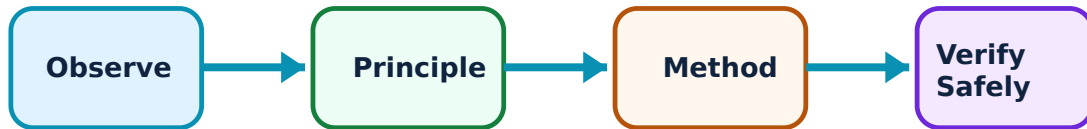
Identify core terms, explain the principle and complete the stated application or practical outcome.

## Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

## Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Drone Hardware and Components: identify the system, state its principle, follow the correct method and verify the result safely.

### 1. Flight Controller and Sensors–

The Flight Controller (FC) is the 'brain' of the drone. It uses an IMU (Inertial Measurement Unit) containing accelerometers and gyroscopes to maintain stability. GPS provides position data, and a Barometer measures altitude. The FC processes sensor data to send commands to the ESCs.

#### Study and application method

Sketch the 'Block Diagram' of a drone's electronic system; explain the role of the 'IMU' in flight stabilization.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

#### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

## How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** Sketch the 'Block Diagram' of a drone's electronic system; explain the role of the 'IMU' in flight stabilization.

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

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**Common mistake / precaution:** Sensor calibration is critical. Do not attempt flight without authorised 'Level' and 'Compass' calibration to prevent erratic behavior or 'flyaways'.

## 2. Propulsion and Power–

BLDC (Brushless DC) motors provide high power-to-weight ratios. ESCs control motor speed. LiPo (Lithium Polymer) batteries are preferred for their high energy density. Propeller selection (diameter and pitch) must match the motor and payload requirements.

### Study and application method

Explain the function of an 'Electronic Speed Controller' (ESC); list three safety precautions for handling 'LiPo Batteries'.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

## How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** Explain the function of an 'Electronic Speed Controller' (ESC); list three safety precautions for handling 'LiPo Batteries'.

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**Common mistake / precaution:** LiPo batteries are a fire hazard if damaged or overcharged. All battery handling must follow authorised charging, storage, and disposal protocols.

**Module 2 revision cue:** Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

## MODULE 3

# Flight Dynamics and Control

6-DOF Dynamics: Roll, Pitch, Yaw; Control algorithms: PID control basics; Wireless communication: Radio control (RC), Telemetry links; FPV (First Person View) systems.

Approx. 16 hours

CO3

Understand · Apply

## Learning goals

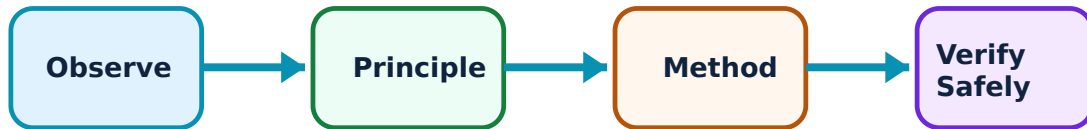
Identify core terms, explain the principle and complete the stated application or practical outcome.

## Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

## Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Flight Dynamics and Control: identify the system, state its principle, follow the correct method and verify the result safely.

## 1. Flight Dynamics and PID Control–

Drones operate in 6 Degrees of Freedom (6-DOF). PID (Proportional-Integral-Derivative) controllers are used to maintain the desired attitude by adjusting motor speeds based on error signals. Tuning PID values is essential for smooth and stable flight.

### Study and application method

Define 'Roll', 'Pitch', and 'Yaw' in drone dynamics; explain the purpose of 'PID Tuning' conceptually.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

## How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** Define 'Roll', 'Pitch', and 'Yaw' in drone dynamics; explain the purpose of 'PID Tuning' conceptually.

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

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**Common mistake / precaution:** Incorrect PID values can cause oscillations or crashes. Do not perform high-speed maneuvers without authorised verification of control stability.

## 2. Communication and FPV–

Radio control (RC) links allow the pilot to command the drone. Telemetry provides real-time data (battery, altitude) back to the Ground Control Station (GCS). FPV systems use a camera and video transmitter (VTX) to provide a pilot's-eye view.

### Study and application method

Describe the role of 'Telemetry' in drone operations; explain the concept of 'First Person View' (FPV).

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

## How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** Describe the role of 'Telemetry' in drone operations; explain the concept of 'First Person View' (FPV).

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

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**Common mistake / precaution:** Signal loss can lead to loss of control. All flights must follow authorised 'Fail-safe' configurations (e.g., Return-to-Home) in case of RC signal loss.

**Module 3 revision cue:** Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

## MODULE 4

# Regulations, Safety and Maintenance

DGCA Regulations: Digital Sky platform, UIN, Remote Pilot License; Airspace zones: Green, Yellow, Red; Pre-flight and Post-flight checklists; Maintenance and Troubleshooting.

Approx. 14 hours

CO4

Understand · Apply

## Learning goals

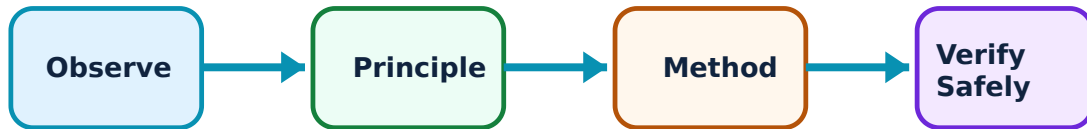
Identify core terms, explain the principle and complete the stated application or practical outcome.

## Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

## Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Regulations, Safety and Maintenance: identify the system, state its principle, follow the correct method and verify the result safely.

### 1. DGCA Regulations and Licensing–

In India, the DGCA regulates drone operations via the 'Digital Sky' platform. Drones (except Nano) require a Unique Identification Number (UIN). Pilots must have a Remote Pilot License for commercial operations. Operations are restricted in 'Yellow' and 'Red' zones.

#### Study and application method

Explain the 'Digital Sky' platform conceptually; list the requirements for obtaining a 'Remote Pilot License'.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

#### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.



## How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** Explain the 'Digital Sky' platform conceptually; list the requirements for obtaining a 'Remote Pilot License'.

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

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**Common mistake / precaution:** Flying in 'No-Fly Zones' (Red zones) is a serious offense. Always check the authorised airspace map on Digital Sky before every flight.

## 2. Safety and Maintenance–

Safety protocols include pre-flight checks (battery, propellers, GPS lock) and maintaining 'Visual Line of Sight' (VLOS). Regular maintenance involves inspecting motors, cleaning sensors, and checking for frame cracks. Troubleshooting involves analyzing flight logs.

### Study and application method

Develop a 'Pre-flight Checklist' conceptually; list three common drone maintenance tasks.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

## How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** Develop a 'Pre-flight Checklist' conceptually; list three common drone maintenance tasks.

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

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**Common mistake / precaution:** Mechanical failure in flight can cause injury or property damage. Do not skip pre-flight inspections; ensure all maintenance follows authorised manufacturer guidelines.

**Module 4 revision cue:** Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

## FORMULA AND PROCEDURE BANK

### Rapid Recall Bank

#### Recall 1

State symbols and SI units before substitution.

#### Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

### Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

### Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

### Recall 5

For a comparison: use construction, working, result, advantage and limitation.

### Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

#### DIAGRAM LIBRARY

## Diagram Library for Rapid Revision

### Module 1: Introduction to UAVs and Aerodynamics

Use the concept flow to revise observation, principle, method and verification.

### Module 2: Drone Hardware and Components

Use the concept flow to revise observation, principle, method and verification.

### Module 3: Flight Dynamics and Control

Use the concept flow to revise observation, principle, method and verification.

## Module 4: Regulations, Safety and Maintenance

Use the concept flow to revise observation, principle, method and verification.

### SELF-LEARNING

## Official Self-Learning and Student Activities

### Structured activity

Compare Multi-rotor and Fixed-wing drones in a conceptual table showing three differences.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

### Structured activity

Sketch the 'Motor Layout' and rotation directions for a standard X-configuration Quadcopter.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

### Structured activity

Draw the 'Airspace Zone' diagram showing Green, Yellow, and Red zones conceptually.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

### Structured activity

Calculate the 'Flight Time' conceptually for a given battery capacity and average current draw.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

## Structured activity

Identify a drone application in your environment (e.g., photography, delivery, agriculture) and note its benefits.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

**Activity discipline:** The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

## EXAMINATION READINESS

# Assessment Methodology and Examination Pattern

### Official assessment structure

| Component                | Official mark / conversion | Student evidence                                      |
|--------------------------|----------------------------|-------------------------------------------------------|
| Attendance               | 5 marks                    | End-of-semester attendance component.                 |
| Self-learning activities | 15 marks                   | Module-linked activities.                             |
| Series examinations      | 20 + 20 converted          | Two 50-mark series examinations.                      |
| CIE total                | Theory CIE 40              | Continuous internal evaluation.                       |
| Written ESE              | Theory ESE 60              | 2.5 hours; official Part A / Part B / Part C pattern. |

## Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

## Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

## EXAM PRACTICE

# Module-wise Questions with Answers

## Module 1 — Introduction to UAVs and Aerodynamics

**Explain Drone Fundamentals and state one correct method, application or precaution.**

A Drone or Unmanned Aerial Vehicle (UAV) is an aircraft without a human pilot on board. Drones are classified by weight (e.g., Nano < 250g) and configuration. Multi-rotors (Quadcopters) are popular for their VTOL (Vertical Take-Off and Landing) capability, while Fixed-wing drones offer better efficiency for long-range tasks.

**Answer framework:** List the 'DGCA Weight Classifications' for drones; explain the four 'Aerodynamic Forces' acting on a drone in flight.

**Important point:** Improper classification can lead to legal violations. Do not operate drones without authorised understanding of weight limits and corresponding regulatory requirements.

**Explain Aerodynamics and Flight Basics and state one correct method, application or precaution.**

Lift is generated by pressure differences (Bernoulli's principle). Weight is the force of gravity. Thrust is provided by motors/propellers. Drag is air resistance. Stable flight requires a balance of these forces. Control is achieved by varying motor speeds in multi-rotors.

**Answer framework:** Describe how a 'Quadcopter' achieves 'Yaw' and 'Pitch' movements conceptually; compare 'Multi-rotor' and 'Fixed-wing' drones.

**Important point:** Aerodynamic instability can cause crashes. Do not fly in high-wind conditions without authorised analysis of the drone's wind-resistance rating.

## Module 2 — Drone Hardware and Components

**Explain Flight Controller and Sensors and state one correct method, application or precaution.**

The Flight Controller (FC) is the 'brain' of the drone. It uses an IMU (Inertial Measurement Unit) containing accelerometers and gyroscopes to maintain stability. GPS provides position data, and a Barometer measures altitude. The FC processes sensor data to send commands to the ESCs.

**Answer framework:** Sketch the 'Block Diagram' of a drone's electronic system; explain the role of the 'IMU' in flight stabilization.

**Important point:** Sensor calibration is critical. Do not attempt flight without authorised 'Level' and 'Compass' calibration to prevent erratic behavior or 'flyaways'.

**Explain Propulsion and Power and state one correct method, application or precaution. –**

BLDC (Brushless DC) motors provide high power-to-weight ratios. ESCs control motor speed. LiPo (Lithium Polymer) batteries are preferred for their high energy density. Propeller selection (diameter and pitch) must match the motor and payload requirements.

**Answer framework:** Explain the function of an 'Electronic Speed Controller' (ESC); list three safety precautions for handling 'LiPo Batteries'.

**Important point:** LiPo batteries are a fire hazard if damaged or overcharged. All battery handling must follow authorised charging, storage, and disposal protocols.

## Module 3 — Flight Dynamics and Control

**Explain Flight Dynamics and PID Control and state one correct method, application or precaution. –**

Drones operate in 6 Degrees of Freedom (6-DOF). PID (Proportional-Integral-Derivative) controllers are used to maintain the desired attitude by adjusting motor speeds based on error signals. Tuning PID values is essential for smooth and stable flight.

**Answer framework:** Define 'Roll', 'Pitch', and 'Yaw' in drone dynamics; explain the purpose of 'PID Tuning' conceptually.

**Important point:** Incorrect PID values can cause oscillations or crashes. Do not perform high-speed maneuvers without authorised verification of control stability.

**Explain Communication and FPV and state one correct method, application or precaution. –**

Radio control (RC) links allow the pilot to command the drone. Telemetry provides real-time data (battery, altitude) back to the Ground Control Station (GCS). FPV systems use a camera and video transmitter (VTX) to provide a pilot's-eye view.

**Answer framework:** Describe the role of 'Telemetry' in drone operations; explain the concept of 'First Person View' (FPV).

**Important point:** Signal loss can lead to loss of control. All flights must follow authorised 'Fail-safe' configurations (e.g., Return-to-Home) in case of RC signal loss.

## Module 4 — Regulations, Safety and Maintenance

**Explain DGCA Regulations and Licensing and state one correct method, application or precaution. –**

In India, the DGCA regulates drone operations via the 'Digital Sky' platform. Drones (except Nano) require a Unique Identification Number (UIN). Pilots must have a Remote Pilot License for commercial operations. Operations are restricted in 'Yellow' and 'Red' zones.

**Answer framework:** Explain the 'Digital Sky' platform conceptually; list the requirements for obtaining a 'Remote Pilot License'.

**Important point:** Flying in 'No-Fly Zones' (Red zones) is a serious offense. Always check the authorised airspace map on Digital Sky before every flight.

**Explain Safety and Maintenance and state one correct method, application or precaution. –**

Safety protocols include pre-flight checks (battery, propellers, GPS lock) and maintaining 'Visual Line of Sight' (VLOS). Regular maintenance involves inspecting motors, cleaning sensors, and checking for frame cracks. Troubleshooting involves analyzing flight logs.

**Answer framework:** Develop a 'Pre-flight Checklist' conceptually; list three common drone maintenance tasks.

**Important point:** Mechanical failure in flight can cause injury or property damage. Do not skip pre-flight inspections; ensure all maintenance follows authorised manufacturer guidelines.

### PRACTICE ONLY

## Practice Model Paper

**Important:** This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTTR model question paper.

### Part A — short response

1. State one key definition from Module 1: Introduction to UAVs and Aerodynamics.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Drone Hardware and Components.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Flight Dynamics and Control.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Regulations, Safety and Maintenance.



8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

### **Part B — structured explanation**

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

### **Part C — extended response / practical task**

1. Develop a complete answer or practical plan for: Definition of UAV/Drone; History and Evolution; Classifications: Nano, Micro, Small, Medium, Large; Aerodynamics: Lift, Weight, Thrust, Drag; Multi-rotor vs Fixed-wing..
2. Develop a complete answer or practical plan for: Flight Controller (FC): Architecture and Sensors (IMU, Barometer, GPS); Electronic Speed Controllers (ESC); BLDC Motors and Propellers; Battery technology (LiPo); Frame materials..
3. Develop a complete answer or practical plan for: 6-DOF Dynamics: Roll, Pitch, Yaw; Control algorithms: PID control basics; Wireless communication: Radio control (RC), Telemetry links; FPV (First Person View) systems..
4. Develop a complete answer or practical plan for: DGCA Regulations: Digital Sky platform, UIN, Remote Pilot License; Airspace zones: Green, Yellow, Red; Pre-flight and Post-flight checklists; Maintenance and Troubleshooting..

#### **LAST-MILE REVISION**

### **Quick Revision**

#### **Module 1: Introduction to UAVs and Aerodynamics**

Definition of UAV/Drone; History and Evolution; Classifications: Nano, Micro, Small, Medium, Large; Aerodynamics: Lift, Weight, Thrust, Drag; Multi-rotor vs Fixed-wing.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

#### **Module 2: Drone Hardware and Components**

Flight Controller (FC): Architecture and Sensors (IMU, Barometer, GPS); Electronic Speed Controllers (ESC); BLDC Motors and Propellers; Battery technology (LiPo); Frame materials.

- Match each topic to CO2.

- Redraw or rehearse one key method.
- Check one common mistake/precaution.

### Module 3: Flight Dynamics and Control

6-DOF Dynamics: Roll, Pitch, Yaw; Control algorithms: PID control basics; Wireless communication: Radio control (RC), Telemetry links; FPV (First Person View) systems.

- Match each topic to C03.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

### Module 4: Regulations, Safety and Maintenance

DGCA Regulations: Digital Sky platform, UIN, Remote Pilot License; Airspace zones: Green, Yellow, Red; Pre-flight and Post-flight checklists; Maintenance and Troubleshooting.

- Match each topic to C04.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

**Common mistakes:** Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

**Exam checklist:** Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

#### VOCABULARY

### Glossary

## Important terms from the syllabus

| Term / topic                    | One-line meaning                                                                                                          |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Drone Fundamentals              | A Drone or Unmanned Aerial Vehicle (UAV) is an aircraft without a human pilot on board.                                   |
| Aerodynamics and Flight Basics  | Lift is generated by pressure differences (Bernoulli's principle).                                                        |
| Flight Controller and Sensors   | The Flight Controller (FC) is the 'brain' of the drone.                                                                   |
| Propulsion and Power            | BLDC (Brushless DC) motors provide high power-to-weight ratios.                                                           |
| Flight Dynamics and PID Control | Drones operate in 6 Degrees of Freedom (6-DOF).                                                                           |
| Communication and FPV           | Radio control (RC) links allow the pilot to command the drone.                                                            |
| DGCA Regulations and Licensing  | In India, the DGCA regulates drone operations via the 'Digital Sky' platform.                                             |
| Safety and Maintenance          | Safety protocols include pre-flight checks (battery, propellers, GPS lock) and maintaining 'Visual Line of Sight' (VLOS). |

## LEARNING RESOURCES

### References

#### Recommended drone technology references

1. Terry Kilby and Belinda Kilby, Getting Started with Drones.
2. Baichtal John, Building Your Own Drones: A Beginner's Guide to Drones, UAVs, and ROVs.
3. K. P. Valavanis and G. J. Vachtsevanos, Handbook of Unmanned Aerial Vehicles.
4. DGCA India, Drone Rules 2021 and Amendments.

#### Approved public drone technology sources

- <https://nptel.ac.in/courses/101108661>
- [https://onlinecourses.nptel.ac.in/noc26\\_ae27/preview](https://onlinecourses.nptel.ac.in/noc26_ae27/preview)

These sources provide the verified study basis while official Course 6044C detail is unavailable; they do not replace official aviation regulations, authorised pilot training, certified equipment specifications or authorised strategic aerial plans.